Deep-learning pansharpening

Launcher: `Toolbox/pansharpening_cnn.bat`

The downloaded application includes the compatible `env_pnn` environment and modified Z-PNN repository. Decompress `env_pnn.7z` as described in the main README before launching the tool; no separate Conda installation is required.

Purpose

The tool aligns matching multispectral and panchromatic rasters, runs one of six Z-PNN-family networks, and exports a georeferenced GeoTIFF.

Inputs and parameters

Option	Meaning and behaviour
Multispectral (MS/ML) raster	WV3/WV2 input must have 8 bands; GE1 must have 4. It must overlap the PAN raster, use the same CRS, and be north-up. Both rasters must hold integer values (raw DN, or reflectance scaled to integers): the output is UInt16, so float reflectance (0–1) is rejected.
Panchromatic (PAN) raster	Matching single-band raster. The common intersection is snapped inward to complete MS pixels, and aligned PAN dimensions are forced to four times the MS width and height.
Pansharpening algorithm	`Z-PNN`, `A-PNN-TA-FR`, `Z-PanNet`, `PanNet-TA-FR`, `Z-DRPNN`, or `DRPNN-TA-FR`. Default: `Z-PNN`. Z-prefixed methods use zero-shot adaptation; TA-FR variants use target-adaptive full-resolution training according to the bundled implementation.
Sensor	`WV3`, `WV2`, or `GE1`. Default: `WV3`. It selects expected band count and pretrained weights; it does not inspect sensor metadata automatically.

Option	Meaning and behaviour
Fine-tuning epochs	Non-negative integer. Default: `1`. `0` runs the loaded pretrained model without fine-tuning. In Z-PNN, `1` is a special shortcut interpreted as 100 epochs for a `Z−` method and 2,000 for a `TA−FR` method.
Use CPU	Off by default. Enable when CUDA is unavailable. Processing will be considerably slower.
Enable co-registration loss	On by default. Adds the bundled model's co-registration term during adaptation. Disable only for an intentional comparison or when troubleshooting that term.
Output folder	Blank creates `pansharp_dl` beside the MS raster. A manually selected folder must already exist.

Outputs

- `<MS-name>_<algorithm_with_underscores>.tif` — georeferenced, tiled, DEFLATE-compressed UInt16 GeoTIFF on the aligned PAN grid. It is always lossless; this tool has no compression option.

GeoTIFF values are clipped to the UInt16 range 0–65535. Temporary aligned rasters, input MAT data and the native Z-PNN MAT result are removed. Source rasters are never modified. Existing outputs may be replaced because this GUI has no overwrite confirmation.

Quality checks

Check alignment on sharp roads and field edges, look for halos or duplicated boundaries, compare each band with the original MS data, and examine output histograms for clipping at 0 or 65535. Network detail is inferred, not directly measured, so do not treat small sharpened forms as independent archaeological evidence.